Arafat Iqbal

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Research Interests

Deep Learning · Computer Vision · Medical Image Analysis · Natural Language Processing · Machine Learning · Environmental Data Science · Precision Agriculture · Human–Computer Interaction

Education

International Islamic University Chittagong Chittagong, Bangladesh
Bachelor of Science in Computer Science and Engineering | ***CGPA: 3.29 / 4.00*** *June 2020 – January 2025*

Publications & Manuscripts Under Review

Privacy-Preserving Federated Continual Learning for Multi-Domain Medical Imaging Under Non-IID Skew
Manuscript Under Review *2026*
- Formulated an end-to-end privacy-preserving federated continual learning framework to mitigate catastrophic forgetting and severe inter-site domain shifts in distributed clinical environments without raw patient data transfer.
- Evaluated systematically across multi-center medical imaging benchmarks under non-IID client partitions, achieving resilient convergence, robust diagnostic stability, and superior cross-domain generalization.

PIDM: A Middleware Framework for Detecting Prompt Injection Attacks in Multi-Agent LLM Pipelines
Manuscript Under Review *2026*
- Architected a modular middleware security layer that intercepts, inspects, and neutralizes direct and indirect prompt injection attacks across inter-agent LLM communication pipelines.
- Developed real-time payload sanitization and semantic policy-checking mechanisms, safeguarding multi-agent collaboration and multi-step tool execution with minimal inference latency.

Research Experience

Research Assistant March 2026 – Present
Dept. of CSE, International Islamic University Chittagong *Chittagong, Bangladesh*
- Conducting computational research under the supervision of Dr. Mohammed Mahmudur Rahman, contributing to ongoing departmental machine learning and computer vision projects.
- Engaging in dataset curation, comprehensive literature reviews, experimental design, and quantitative benchmark evaluation across clinical imaging and environmental monitoring domains.

Professional Experience

Software Engineer February 2025 – Present
Craftsmen IT *Dhaka, Bangladesh*
- Engineered full-stack SaaS solutions, cloud architectures, and scalable APIs adhering to clean architecture.
- Collaborated across engineering teams to implement scalable features, automated testing, and CI/CD pipelines.

Undergraduate Teaching Assistant June 2022 – December 2024
International Islamic University Chittagong *Chittagong, Bangladesh*
- Mentored undergraduate students across core computer science courses, conducting discussion and problem-solving clinics.
- Evaluated laboratory assignments and course projects, delivering actionable feedback to strengthen algorithmic understanding.

Selected Projects

Alainstar ERP & E-Commerce January 2026 – Present

Next.js, Laravel, PostgreSQL, Docker, CI/CD, AWS Services

- Engineered a custom enterprise ERP and e-commerce platform for a UAE-based auto parts dealer, automating inventory tracking and order management.
- Optimized catalog lookup REST APIs and checkout workflows, eliminating operational bottlenecks and reducing query latency.

freemail.ai May 2025 – July 2026

Python, Postfix, Redis, REST API

- Architected a disposable temporary email platform featuring dynamic multi-domain SMTP routing and auto-purge storage.
- Engineered high-speed REST APIs handling real-time high-concurrency web traffic.

Bengali Isolated Number Recognition Using Air-Writing Techniques November 2024

Undergraduate Thesis | Python, OpenCV, CNN, AlexNet

- Engineered a camera-based computer vision framework for recognizing Bengali air-written digits via real-time hand-motion tracking and trajectory-to-image conversion, achieving 98% accuracy on 85,000 samples.

Brain Tumor Detection and Classification on MRI Scans Manuscript in Preparation

Deep Learning | EfficientNetV4, DenseNet, MobileNetV2, Grad-CAM

- Developed an automated multi-class classification pipeline on brain MRI scans, achieving up to 99% accuracy with a DenseNet ensemble and Grad-CAM interpretability visualizations.

Technical Skills

Programming & Frameworks: Python (TensorFlow, Keras, PyTorch, Scikit-learn, Pandas, NumPy), SQL (MySQL, PostgreSQL), Next.js, Laravel

Developer Tools & Cloud: Docker, AWS Services, CI/CD, Git, GitHub, Linux, Postfix, Redis, LaTeX, Postman

Visualization & Management: Tableau, Power BI, Figma, Jira, Trello, Agile/Scrum

Leadership & Extracurricular Activities

IEEE RAS Robotics and Automation Society (IIUC) Chittagong, Bangladesh

General Secretary | Former Membership Coordinator

- Directed society operations, arranged technical robotics workshops, and spearheaded competitive technical events.

IIUC Computer Club & Data Science Community Chittagong, Bangladesh

Event Secretary, Computer Club | Executive Member, Data Science Community

- Organized intra-university programming contests, data science seminars, and hands-on ML workshops.